

# Bryan Pham

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## EDUCATION

### University of California, Los Angeles (UCLA)

B.S. Degree: Mechanical Engineering (Transfer GPA: 4.0)

Los Angeles, CA

June. 2026 – May 2028

## EXPERIENCE

### Design & Manufacturing Engineering Intern

June. 2026 – Present

Mission College | <https://abgscce-website.vercel.app/>

Santa Clara, CA

- Designed and optimized the airframe for a high-powered model rocket in Onshape and SolidWorks, including a tilt and roll control system, ensuring structural integrity, steady component implementation, and at least 1.0 cal of stability in the vehicle
- Conducted 3000+ hours of computational fluid dynamics (CFD) simulations in SimScale to validate aerodynamic performance, drag, pressure distribution, and stability prior to fabrication
- Engineered an active-control tilt/roll system using micro-servos, reading acceleration and angular velocity data from an IMU module, and translating changes to canard deflections with an ESP32 and Raspberry Pi 5 system
- Prototyped and implemented live video capture with a drone runcam to analyze interactive differences between CFD simulation and live runs while reducing drag

### Associated Student Government Senator

Aug. 2025 – May 2026

Mission College

Santa Clara, CA

- Represented 6,000+ students in recurring discussions with college administrators and campus leadership, surfacing student concerns into actionable policy proposals
- Advocated for policies & programs that improve student engagement, academic success, and resource accessibility
- Selected to represent student government at an institutional leadership retreat, presenting academic and operational recommendations directly to district executives

## PROJECTS

### Aside AI | 1st DeepGram Track, UC Berkeley AI Hackathon | DeepGram, Redis, QNX, Python, Microcomputers

- Built a clip-on camera and mic that narrates the user's surroundings in real time through different AI personalities
- Developed a modular system using a Raspberry Pi (QNX 8.0) capture device, Python coordination, and React Native app, achieving live end-to-end narration in 1–2 seconds

### Mission Launch Rocketry | OnShape, 3D Printing, Microcomputers, Microcontrollers

- Founded and served as President of college rocketry club, overseeing 52 members, managing budget, coordinating project/promotional events, and directing design-build-launch projects from concept through flight
- Integrated dual-deployment recovery system in a two-stage high power rocket (drogue and main parachutes) with staged separation sequencing from EasyMini and EasyMega flight computers
- Designed and implemented a minimum diameter airframe for a maximum velocity model rocket, containing an L2 class motor, achieving a simulated top speed of Mach 2.6

### Recco | Y Combinator AI Growth Hackathon | Computer Vision, Deepgram, Fiber, Convex, CV Service

- Built a camera-first iOS networking assistant that identifies people at events in real time by fusing on-device face tracking with cloud vision, identity lookup, and voice control
- Engineered a SwiftUI + AVFoundation camera pipeline with Apple Vision face tracking, a target-lock reticle, and an AR overlay that resolves the person nearest screen center on voice or text command

### RollAway | 1st Beginner Track, MLH x DigitalOcean Hackathon | React, TypeScript, FastAPI

- A map-first location-intelligence and permit-planning app for mobile food vendors in San Francisco that ranks legal, low-competition places to set up for a chosen time window and simplifies the permit acquisition process
- Built the permit checklist functionality using React and TypeScript on a DigitalOcean serverless backend and implemented a Zustand-driven UI that consolidates user-ingested data into auto-filled forms with manual text entry to improve accessibility

### Camera Tracking Drone | Microcomputers, Microcontrollers, Python, Computer Vision

- Designing toward a sub-250g autonomous tracking drone that follows only knowingly registered users, starting with a validated desktop tracking prototype
- Built a modular real-time tracking pipeline in Python (CUDA-accelerated YOLO11n detection, ByteTrack for persistent IDs, click-to-select targeting emitting a normalized frame-center error signal), configured via a typed YAML config with env overrides and covered by pytest unit and integration tests

## TECHNICAL SKILLS

**Languages :** C, C++, Python, TypeScript, JavaScript, MATLAB, Swift, HTML/CSS

**CAD:** SolidWorks, Onshape, AutoCAD, Fusion360, MecAgent, Zoo.dev

**Manufacturing:** 3D Printing, Soldering, Microcontrollers, SimScale, Microcomputers, Computer Vision, PSpice, LTSpice, Oscilloscope, Function Generator

**Tools & Platforms:** Claude Code, Codex, Cursor, Git/Github, Github Actions, Redis, Mapbox